

CHL5233: Quarto in-class assignment

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For this in-class assignment, you will work in a group to create your **individual** Quarto document. Please consult your classmates and [Tutorial: Authoring](#) webpage by [Quarto](#). You can also use the Internet including ChatGPT. The TA and I are also available for any questions.

When you are finished, render the document to PDF. Please carefully **proof read** and make edits as needed. **Commit** and **push** the edits to this file and the new PDF file to your **new repository**.

Create a new repository

- Under your GitHub account, create a **new** repository called `armed_conflict`
 - Throughout the course, we will use this repository to **reproduce** and **replicate** the analysis in the paper
- Create **two** folders
 - `data`
 - `reports`
- Save `countries.txt` and `regions.txt` into `data/raw/` subfolder
- Save `quarto_inclass.qmd` into `reports/` folder
- Eventually, your repository will have a very organized structure like this:

Example of data science organization

```
armed_conflict/  
  README.md  
  data/  
    raw/  
    processed/  
  scripts/  
    01_import.R  
    02_clean.R
```

```
03_analysis.R
outputs/
  figures/
  tables/
reports/
  analysis.qmd
```

Edit this file to create a PDF document using Quarto

The R code chunk below includes the R packages required to knit this document. This is often called a **setup** code. We want the code to run, but we don't want the final document to show the code or results. Edit the code chunk so that the code will run but the code and results (including all warnings and messages) are hidden from the final PDF output.

For this exercise, we will use the following two data sets.

1. **countries.txt**: A data set that contains a list of countries used in the paper “Implications of armed conflict for maternal and child health: A regression analysis of data from 181 countries for 2000–2019” and their three-digit ISO (International Organization for Standardization) code.
2. **regions.txt**: A data set that contains a list of all major countries in the world and their
 - a. ISO code (**country_code**)
 - b. region (**region**)
 - c. sub-region (**sub_region**)
 - d. numerical region code (**region_code**)
 - e. numerical sub-region code (**sub_region_code**)

The **final goal** is to display a **bar chart** that depicts how many countries are in each region.

Read in data and create an analytical data set

First, edit the code chunk below that reads in **countries.txt** to also display the number of rows and columns of the data.

```
countries <- read.table(
  file="/Users/owner/Desktop/Statistical Programming/armed_conflict/armed_conflict/data/raw/count
dim(countries)
```

```
[1] 186  2
```

186 countries in the data set

Second, read in `regions.txt` and also write a sentence that states how many countries are included in the data set.

```
regions<- read.table(file="/Users/owner/Desktop/Statistical Programming/armed_conflict/armed_conflict.txt")
dim(regions)
```

```
[1] 249  6
```

249 countries are added in the dataset

Third, subset the `regions` data set to only include countries from the `countries` data set.

```
filtered <- filter(regions, country_code %in% countries$ISO)
filtered
```

	country_name	country_code	region
1	Afghanistan	AFG	Asia
2	Albania	ALB	Europe
3	Algeria	DZA	Africa
4	Andorra	AND	Europe
5	Angola	AGO	Africa
6	Antigua and Barbuda	ATG	Americas
7	Argentina	ARG	Americas
8	Armenia	ARM	Asia
9	Australia	AUS	Oceania
10	Austria	AUT	Europe
11	Azerbaijan	AZE	Asia
12	Bahamas	BHS	Americas
13	Bahrain	BHR	Asia
14	Bangladesh	BGD	Asia
15	Barbados	BRB	Americas
16	Belarus	BLR	Europe
17	Belgium	BEL	Europe
18	Belize	BLZ	Americas
19	Benin	BEN	Africa
20	Bhutan	BTN	Asia
21	Bolivia (Plurinational State of)	BOL	Americas
22	Bosnia and Herzegovina	BIH	Europe
23	Botswana	BWA	Africa
24	Brazil	BRA	Americas
25	Brunei Darussalam	BRN	Asia

26	Bulgaria	BGR	Europe
27	Burkina Faso	BFA	Africa
28	Burundi	BDI	Africa
29	Cabo Verde	CPV	Africa
30	Cambodia	KHM	Asia
31	Cameroon	CMR	Africa
32	Canada	CAN	Americas
33	Central African Republic	CAF	Africa
34	Chad	TCD	Africa
35	Chile	CHL	Americas
36	China	CHN	Asia
37	Colombia	COL	Americas
38	Comoros	COM	Africa
39	Congo	COG	Africa
40	Congo, Democratic Republic of the	COD	Africa
41	Costa Rica	CRI	Americas
42	Cote d'Ivoire	CIV	Africa
43	Croatia	HRV	Europe
44	Cuba	CUB	Americas
45	Cyprus	CYP	Asia
46	Czechia	CZE	Europe
47	Denmark	DNK	Europe
48	Djibouti	DJI	Africa
49	Dominica	DMA	Americas
50	Dominican Republic	DOM	Americas
51	Ecuador	ECU	Americas
52	Egypt	EGY	Africa
53	El Salvador	SLV	Americas
54	Equatorial Guinea	GNQ	Africa
55	Eritrea	ERI	Africa
56	Estonia	EST	Europe
57	Eswatini	SWZ	Africa
58	Ethiopia	ETH	Africa
59	Fiji	FJI	Oceania
60	Finland	FIN	Europe
61	France	FRA	Europe
62	Gabon	GAB	Africa
63	Gambia	GMB	Africa
64	Georgia	GEO	Asia
65	Germany	DEU	Europe
66	Ghana	GHA	Africa
67	Greece	GRC	Europe
68	Grenada	GRD	Americas
69	Guatemala	GTM	Americas
70	Guinea	GIN	Africa

71	Guinea-Bissau	GNB	Africa
72	Guyana	GUY	Americas
73	Haiti	HTI	Americas
74	Honduras	HND	Americas
75	Hungary	HUN	Europe
76	Iceland	ISL	Europe
77	India	IND	Asia
78	Indonesia	IDN	Asia
79	Iran (Islamic Republic of)	IRN	Asia
80	Iraq	IRQ	Asia
81	Ireland	IRL	Europe
82	Italy	ITA	Europe
83	Jamaica	JAM	Americas
84	Japan	JPN	Asia
85	Jordan	JOR	Asia
86	Kazakhstan	KAZ	Asia
87	Kenya	KEN	Africa
88	Kiribati	KIR	Oceania
89	Korea (Democratic People's Republic of)	PRK	Asia
90	Korea, Republic of	KOR	Asia
91	Kuwait	KWT	Asia
92	Kyrgyzstan	KGZ	Asia
93	Lao People's Democratic Republic	LAO	Asia
94	Latvia	LVA	Europe
95	Lebanon	LBN	Asia
96	Lesotho	LSO	Africa
97	Liberia	LBR	Africa
98	Libya	LBY	Africa
99	Lithuania	LTU	Europe
100	Luxembourg	LUX	Europe
101	Madagascar	MDG	Africa
102	Malawi	MWI	Africa
103	Malaysia	MYS	Asia
104	Maldives	MDV	Asia
105	Mali	MLI	Africa
106	Malta	MLT	Europe
107	Marshall Islands	MHL	Oceania
108	Mauritania	MRT	Africa
109	Mauritius	MUS	Africa
110	Mexico	MEX	Americas
111	Micronesia (Federated States of)	FSM	Oceania
112	Moldova, Republic of	MDA	Europe
113	Mongolia	MNG	Asia
114	Montenegro	MNE	Europe
115	Morocco	MAR	Africa

116	Mozambique	MOZ	Africa
117	Myanmar	MMR	Asia
118	Namibia	NAM	Africa
119	Nepal	NPL	Asia
120	Netherlands	NLD	Europe
121	New Zealand	NZL	Oceania
122	Nicaragua	NIC	Americas
123	Niger	NER	Africa
124	Nigeria	NGA	Africa
125	North Macedonia	MKD	Europe
126	Norway	NOR	Europe
127	Oman	OMN	Asia
128	Pakistan	PAK	Asia
129	Panama	PAN	Americas
130	Papua New Guinea	PNG	Oceania
131	Paraguay	PRY	Americas
132	Peru	PER	Americas
133	Philippines	PHL	Asia
134	Poland	POL	Europe
135	Portugal	PRT	Europe
136	Puerto Rico	PRI	Americas
137	Qatar	QAT	Asia
138	Romania	ROU	Europe
139	Russian Federation	RUS	Europe
140	Rwanda	RWA	Africa
141	Saint Lucia	LCA	Americas
142	Saint Vincent and the Grenadines	VCT	Americas
143	Samoa	WSM	Oceania
144	Sao Tome and Principe	STP	Africa
145	Saudi Arabia	SAU	Asia
146	Senegal	SEN	Africa
147	Serbia	SRB	Europe
148	Seychelles	SYC	Africa
149	Sierra Leone	SLE	Africa
150	Singapore	SGP	Asia
151	Slovakia	SVK	Europe
152	Slovenia	SVN	Europe
153	Solomon Islands	SLB	Oceania
154	Somalia	SOM	Africa
155	South Africa	ZAF	Africa
156	South Sudan	SSD	Africa
157	Spain	ESP	Europe
158	Sri Lanka	LKA	Asia
159	Sudan	SDN	Africa
160	Suriname	SUR	Americas

161	Sweden	SWE	Europe
162	Switzerland	CHE	Europe
163	Syrian Arab Republic	SYR	Asia
164	Tajikistan	TJK	Asia
165	Tanzania, United Republic of	TZA	Africa
166	Thailand	THA	Asia
167	Timor-Leste	TLS	Asia
168	Togo	TGO	Africa
169	Tonga	TON	Oceania
170	Trinidad and Tobago	TTO	Americas
171	Tunisia	TUN	Africa
172	Turkey	TUR	Asia
173	Turkmenistan	TKM	Asia
174	Uganda	UGA	Africa
175	Ukraine	UKR	Europe
176	United Arab Emirates	ARE	Asia
177	United Kingdom of Great Britain and Northern Ireland	GBR	Europe
178	United States of America	USA	Americas
179	Uruguay	URY	Americas
180	Uzbekistan	UZB	Asia
181	Vanuatu	VUT	Oceania
182	Venezuela (Bolivarian Republic of)	VEN	Americas
183	Viet Nam	VNM	Asia
184	Yemen	YEM	Asia
185	Zambia	ZMB	Africa
186	Zimbabwe	ZWE	Africa

	sub_region	region_code	sub_region_code
1	Southern Asia	142	34
2	Southern Europe	150	39
3	Northern Africa	2	15
4	Southern Europe	150	39
5	Sub-Saharan Africa	2	202
6	Latin America and the Caribbean	19	419
7	Latin America and the Caribbean	19	419
8	Western Asia	142	145
9	Australia and New Zealand	9	53
10	Western Europe	150	155
11	Western Asia	142	145
12	Latin America and the Caribbean	19	419
13	Western Asia	142	145
14	Southern Asia	142	34
15	Latin America and the Caribbean	19	419
16	Eastern Europe	150	151
17	Western Europe	150	155
18	Latin America and the Caribbean	19	419

19	Sub-Saharan Africa	2	202
20	Southern Asia	142	34
21	Latin America and the Caribbean	19	419
22	Southern Europe	150	39
23	Sub-Saharan Africa	2	202
24	Latin America and the Caribbean	19	419
25	South-eastern Asia	142	35
26	Eastern Europe	150	151
27	Sub-Saharan Africa	2	202
28	Sub-Saharan Africa	2	202
29	Sub-Saharan Africa	2	202
30	South-eastern Asia	142	35
31	Sub-Saharan Africa	2	202
32	Northern America	19	21
33	Sub-Saharan Africa	2	202
34	Sub-Saharan Africa	2	202
35	Latin America and the Caribbean	19	419
36	Eastern Asia	142	30
37	Latin America and the Caribbean	19	419
38	Sub-Saharan Africa	2	202
39	Sub-Saharan Africa	2	202
40	Sub-Saharan Africa	2	202
41	Latin America and the Caribbean	19	419
42	Sub-Saharan Africa	2	202
43	Southern Europe	150	39
44	Latin America and the Caribbean	19	419
45	Western Asia	142	145
46	Eastern Europe	150	151
47	Northern Europe	150	154
48	Sub-Saharan Africa	2	202
49	Latin America and the Caribbean	19	419
50	Latin America and the Caribbean	19	419
51	Latin America and the Caribbean	19	419
52	Northern Africa	2	15
53	Latin America and the Caribbean	19	419
54	Sub-Saharan Africa	2	202
55	Sub-Saharan Africa	2	202
56	Northern Europe	150	154
57	Sub-Saharan Africa	2	202
58	Sub-Saharan Africa	2	202
59	Melanesia	9	54
60	Northern Europe	150	154
61	Western Europe	150	155
62	Sub-Saharan Africa	2	202
63	Sub-Saharan Africa	2	202

64	Western Asia	142	145
65	Western Europe	150	155
66	Sub-Saharan Africa	2	202
67	Southern Europe	150	39
68	Latin America and the Caribbean	19	419
69	Latin America and the Caribbean	19	419
70	Sub-Saharan Africa	2	202
71	Sub-Saharan Africa	2	202
72	Latin America and the Caribbean	19	419
73	Latin America and the Caribbean	19	419
74	Latin America and the Caribbean	19	419
75	Eastern Europe	150	151
76	Northern Europe	150	154
77	Southern Asia	142	34
78	South-eastern Asia	142	35
79	Southern Asia	142	34
80	Western Asia	142	145
81	Northern Europe	150	154
82	Southern Europe	150	39
83	Latin America and the Caribbean	19	419
84	Eastern Asia	142	30
85	Western Asia	142	145
86	Central Asia	142	143
87	Sub-Saharan Africa	2	202
88	Micronesia	9	57
89	Eastern Asia	142	30
90	Eastern Asia	142	30
91	Western Asia	142	145
92	Central Asia	142	143
93	South-eastern Asia	142	35
94	Northern Europe	150	154
95	Western Asia	142	145
96	Sub-Saharan Africa	2	202
97	Sub-Saharan Africa	2	202
98	Northern Africa	2	15
99	Northern Europe	150	154
100	Western Europe	150	155
101	Sub-Saharan Africa	2	202
102	Sub-Saharan Africa	2	202
103	South-eastern Asia	142	35
104	Southern Asia	142	34
105	Sub-Saharan Africa	2	202
106	Southern Europe	150	39
107	Micronesia	9	57
108	Sub-Saharan Africa	2	202

109	Sub-Saharan Africa	2	202
110	Latin America and the Caribbean	19	419
111	Micronesia	9	57
112	Eastern Europe	150	151
113	Eastern Asia	142	30
114	Southern Europe	150	39
115	Northern Africa	2	15
116	Sub-Saharan Africa	2	202
117	South-eastern Asia	142	35
118	Sub-Saharan Africa	2	202
119	Southern Asia	142	34
120	Western Europe	150	155
121	Australia and New Zealand	9	53
122	Latin America and the Caribbean	19	419
123	Sub-Saharan Africa	2	202
124	Sub-Saharan Africa	2	202
125	Southern Europe	150	39
126	Northern Europe	150	154
127	Western Asia	142	145
128	Southern Asia	142	34
129	Latin America and the Caribbean	19	419
130	Melanesia	9	54
131	Latin America and the Caribbean	19	419
132	Latin America and the Caribbean	19	419
133	South-eastern Asia	142	35
134	Eastern Europe	150	151
135	Southern Europe	150	39
136	Latin America and the Caribbean	19	419
137	Western Asia	142	145
138	Eastern Europe	150	151
139	Eastern Europe	150	151
140	Sub-Saharan Africa	2	202
141	Latin America and the Caribbean	19	419
142	Latin America and the Caribbean	19	419
143	Polynesia	9	61
144	Sub-Saharan Africa	2	202
145	Western Asia	142	145
146	Sub-Saharan Africa	2	202
147	Southern Europe	150	39
148	Sub-Saharan Africa	2	202
149	Sub-Saharan Africa	2	202
150	South-eastern Asia	142	35
151	Eastern Europe	150	151
152	Southern Europe	150	39
153	Melanesia	9	54

154	Sub-Saharan Africa	2	202
155	Sub-Saharan Africa	2	202
156	Sub-Saharan Africa	2	202
157	Southern Europe	150	39
158	Southern Asia	142	34
159	Northern Africa	2	15
160	Latin America and the Caribbean	19	419
161	Northern Europe	150	154
162	Western Europe	150	155
163	Western Asia	142	145
164	Central Asia	142	143
165	Sub-Saharan Africa	2	202
166	South-eastern Asia	142	35
167	South-eastern Asia	142	35
168	Sub-Saharan Africa	2	202
169	Polynesia	9	61
170	Latin America and the Caribbean	19	419
171	Northern Africa	2	15
172	Western Asia	142	145
173	Central Asia	142	143
174	Sub-Saharan Africa	2	202
175	Eastern Europe	150	151
176	Western Asia	142	145
177	Northern Europe	150	154
178	Northern America	19	21
179	Latin America and the Caribbean	19	419
180	Central Asia	142	143
181	Melanesia	9	54
182	Latin America and the Caribbean	19	419
183	South-eastern Asia	142	35
184	Western Asia	142	145
185	Sub-Saharan Africa	2	202
186	Sub-Saharan Africa	2	202

Tabulate countries by region and sub-region

Next, we want to find out how many countries are in a region. Use the `group_by()` and `summarise()` functions in the `dplyr` package to count the number of countries in each region. Then use the `kable()` function in the `knitr` package to create a simple table with the column headers, “Region” and “Count”.

```
countries_in_region<- regions %>%
  mutate(region = if_else(region == "", "Unknown", region)) %>%
  group_by(region) %>%
```

```
summarise(total_count = n())

kable(countries_in_region, col.names = c("Region", "Count"))
```

Region	Count
Africa	60
Americas	57
Asia	51
Europe	51
Oceania	29
Unknown	1

Create a similar table that shows the number of countries in each **sub-region**.

```
countries_in_sub_region<- regions %>%
  mutate(sub_region = if_else(sub_region == "", "Unknown", sub_region)) %>%
  group_by(sub_region) %>%
  summarise(count=length(country_name))

kable(countries_in_sub_region, col.names = c("Sub-Region", "Count"))
```

Sub-Region	Count
Australia and New Zealand	6
Central Asia	5
Eastern Asia	8
Eastern Europe	10
Latin America and the Caribbean	52
Melanesia	5
Micronesia	8
Northern Africa	7
Northern America	5
Northern Europe	16
Polynesia	10
South-eastern Asia	11
Southern Asia	9
Southern Europe	16
Sub-Saharan Africa	53
Unknown	1
Western Asia	18
Western Europe	9

Data visualization using ggplot

Here, we will use `geom_bar()` function within `ggplot()` to create a bar chart that displays the **number of countries** in each **sub-region**.

1. Color code each sub-region by **region**.
2. Create bars that run horizontally where sub-region names appear along the y-axis.
3. Order the bars such that the sub-region with the most number countries appear at the top and the sub-region with the least number of countries appear at the bottom.

```
ggplot(countries_in_sub_region, aes(x= count, y=reorder(sub_region,count), fill=sub_region))+
  geom_bar(stat="identity")+
  labs(
    title = "Number of Countries in Each Sub-Region",
    y= "Total Number of Countries",
    x= "Sub-Region")
```

